Shot Timer MK1: Project Book

Oct 19, 2026 · @Is This Ai Free

At a glance

The Shot Timer MK1 is a finished, working open-source shot timer: firmware v1.7.5 is final and in use, and the hardware documentation is a first release with a short list of wiring details still being confirmed on a built unit. You press START, wait for a random beep, shoot, and press STOP to read your draw time, splits, totals and average on a two-color OLED.



A finished unit: tan lid with the Huckleberry Innovations name, olive green shell, red STOP and green START buttons, and the clip-on lavalier mic.

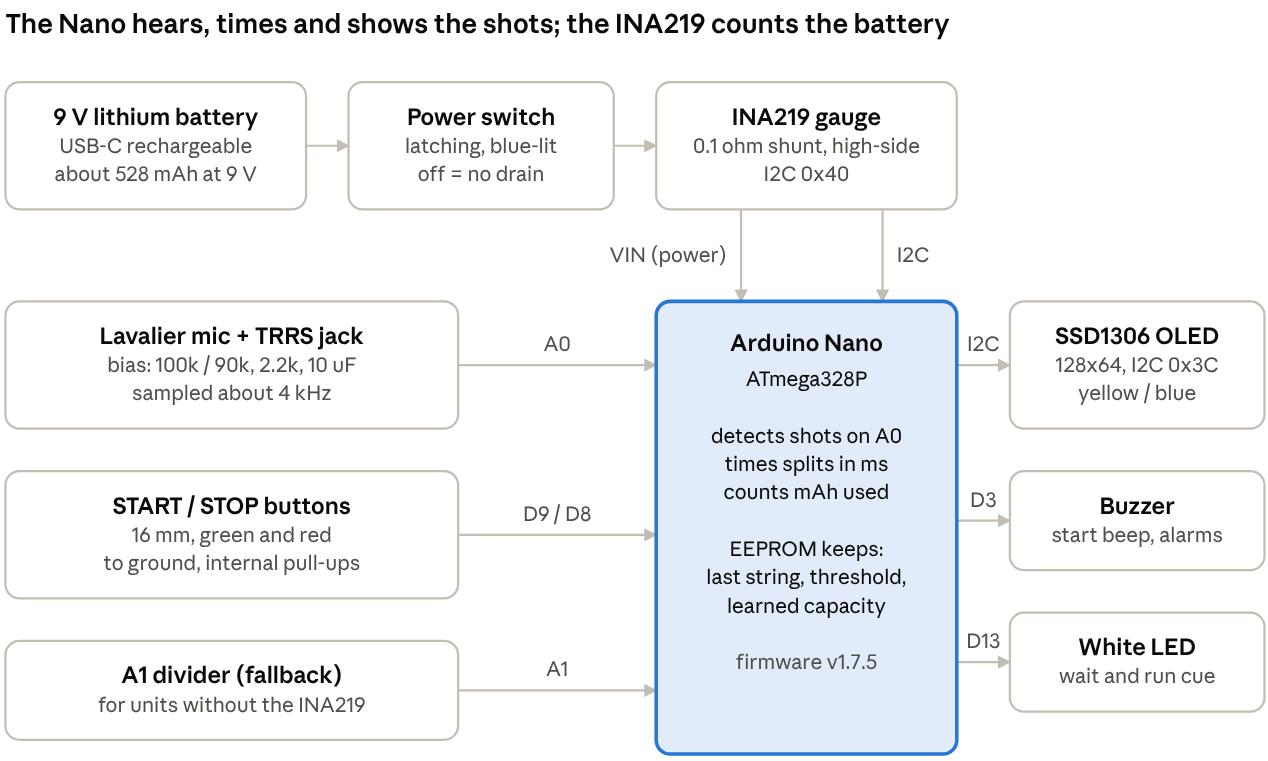
Item	Detail
Maker	Huckleberry Innovations LLC (Josue F. Kure), Lexington, SC
Repository	github.com/jfkure/shot-timer-mk1 (public)
Controller	Arduino Nano (ATmega328P)
Display	0.96 in SSD1306 128x64 OLED, yellow/blue, I2C 0x3C
Shot detection	Clip-on lavalier mic on a 3.5 mm TRRS jack, read on A0
Power	9 V USB-C rechargeable lithium battery only (no alkaline), about 16 hours per charge
Battery gauge	INA219 with 0.1 ohm shunt, high-side in the battery line, I2C 0x40 (added 2026-10-16)
Enclosure	v6, two parts (bottom shell and lid), multi-color PLA+
Firmware	v1.7.5, final

Licenses: firmware GPL-3.0-or-later, hardware CERN-OHL-S-2.0, documentation CC BY-SA 4.0. The names "Huckleberry Innovations" and "Shot Timer MK1" are not covered, so derivative builds must use another name.

Key features: random 3 to 8 s start delay and 0.5 s beep; up to 60 shots per string shown to two decimals; result pages for 1ST SHOT, AVERAGE, SPLITS, TOTALS and BATTERY; last string kept through power-off; adjustable sound threshold (0 to 500, default 150) with a live mic test; battery percentage on every screen with LOW and CHARGE warnings and automatic capacity learning.

How it works

A microphone listens for shots, the Nano times them in milliseconds from a random start beep, and the OLED shows the results; an INA219 counts the battery charge used so the timer can show an honest percentage.



System block diagram · power across the top, inputs left, outputs right

Battery power runs through the switch and the INA219 shunt into the Nano; the mic and buttons feed the Nano, which drives the OLED, buzzer and LED. The A1 divider only matters on early units without the INA219.

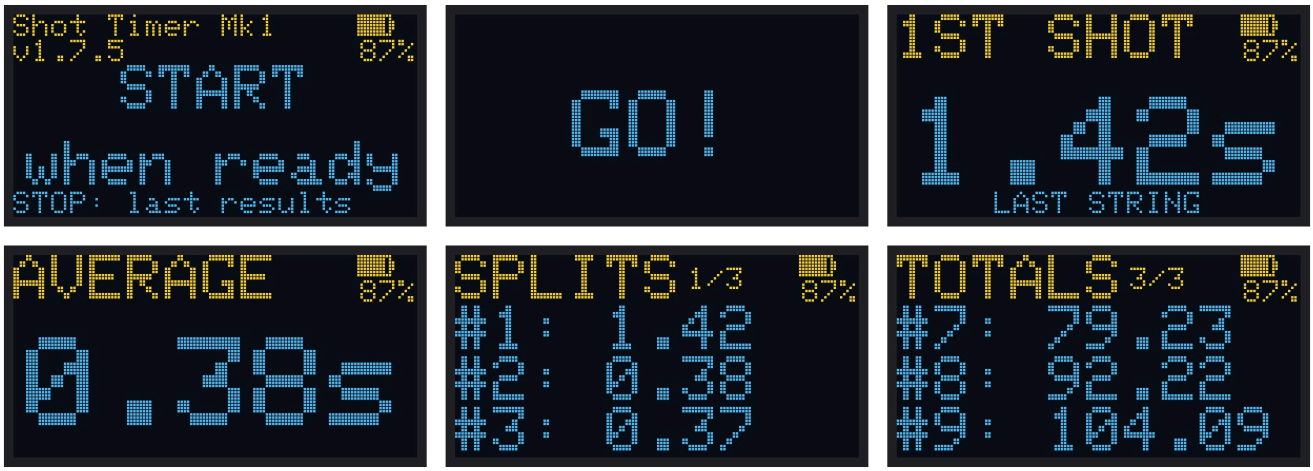
Using it

Two buttons run everything. A **tap** is a quick press; a **hold** is 2 or 3 seconds.

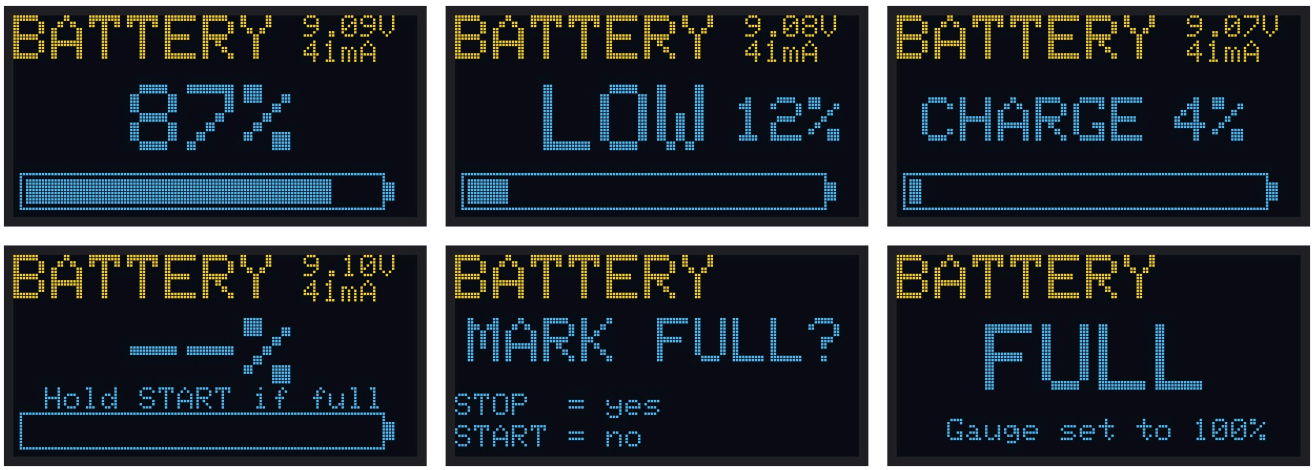
1. Tap START. After a random 3 to 8 s wait the buzzer beeps for 0.5 s, the screen shows GO! and the clock starts. The white LED is on during the wait and blinks while the string runs.
2. Shoot. Each shot louder than the threshold is logged; sounds within 0.10 s of a shot are ignored so echoes and ringing steel don't count.
3. Press STOP. The clock stops at once, the string is saved to EEPROM, and the results appear.
4. Tap STOP to step through 1ST SHOT, AVERAGE, SPLITS, TOTALS and BATTERY, then back to 1ST SHOT.

Button press	Where	What happens
Tap START	Start screen or any result page	Starts a string
Press STOP	During a string	Ends the string and shows results
Tap STOP	Start screen or result pages	Opens results / next page (on release)
Hold START 2 s	Any page except BATTERY	Test mode (live mic level)
Hold START 3 s, then tap STOP	BATTERY page, after a full charge	Marks the battery FULL
Hold STOP 2 s	Start screen or result pages	Sound threshold setting
Hold START + STOP 3 s	Start screen or result pages	Factory reset (keeps learned battery capacity)

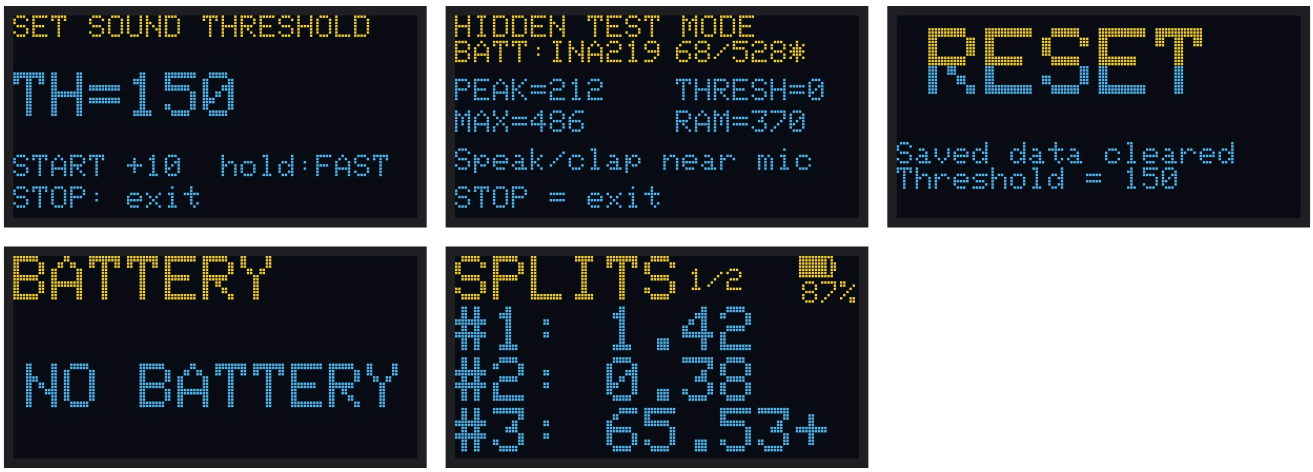
The screens



Shooting: start screen, GO!, 1ST SHOT, AVERAGE, SPLITS and TOTALS, drawn exactly as the v1.7.5 firmware draws them.



Battery: normal, LOW (15% or less), CHARGE (5% or less), not marked yet, MARK FULL? and FULL.



Setup and special cases: sound threshold, test mode, factory reset, NO BATTERY, and a gap over 65.5 s shown as 65.53+.

Battery warnings

Warning	Trigger	BATTERY page	Other pages
None	Above 15% and above 8.5 V	Percentage and fill bar	No beeps
LOW	15% or less, or below 8.5 V	Flashing LOW, beep every 10 s	Beep every 60 s
CHARGE	5% or less, or below 7.9 V	Flashing CHARGE, double beep every 5 s	Double beep every 30 s

Warning	Trigger	BATTERY page	Other pages
NO BATTERY	Below 6.0 V (switched off or USB only)	NO BATTERY	No beeps

Beeps never sound during a string, the start wait, test mode or settings, and pause for 3 minutes after any button press. LOW gives about 2 to 2.5 hours of warning and CHARGE about 45 to 50 minutes, both confirmed in full run-down tests.

Hardware

Everything is hand-wired on a 30-row ElectroCookie protoboard around an Arduino Nano, with the panel parts (buttons, switch, LED, buzzer, jack, battery) on 2-pin connectors so they unplug for service. Items marked **PENDING** below are still to be confirmed on a built unit (see Open items).



The finished board with the OLED mounted on the underside, showing the start screen.

Parts list

Ref	Part	Qty	Pack price (USD)	Notes
U1	Arduino Nano (ATmega328P), USB-C	1	15.99 / 3	Rows 16 to 30
U2	0.96 in SSD1306 OLED, 128x64, yellow/blue, I2C	1	6.99	0x3C, underside of rows 5 to 8
U3	INA219 current sensor, 0.1 ohm shunt	1	9.99 / 3	0x40, added 2026-10-16
PCB1	ElectroCookie solderable protoboard	1	8.79 / 5	
MK1	Clip-on lavalier mic, 3.5 mm TRRS	1	22.99	Was 9.99; check stock
J1	TRRS 3.5 mm breakout	1	10.99 / 6	Mic jack
BT1	9 V USB-C rechargeable lithium battery	1	25.00	PENDING: one battery or a pack?
BT1-CLIP	9 V battery clip with leads	1	6.99 / 10	
SW1	Latching power switch, blue-lit, guarded	1	14.39 / 6	
SW2	16 mm momentary button, green (START)	1	5.75 / 5	D9
SW3	16 mm momentary button, red (STOP)	1	5.75 / 5	D8
LS1	Buzzer	1	8.69 / 5	D3
D1	White LED, panel mount	1	6.99 / 5	D13; PENDING: built-in resistor?
CONN	2-pin wire connector	3	6.99 / 30	Comes in pairs
R1	10k 1%	1	0.24 / 25	Battery divider top

Ref	Part	Qty	Pack price (USD)	Notes
R2a, R2b	2.2k + 4.7k 1% in series (6.9k)	1 each	0.24 / 25	Battery divider bottom
R3	100k 1%	1	0.24 / 25	Mic bias top
R4a, R4b	22k + 68k 1% in series (90k)	1 each	0.24 / 25	Mic bias bottom
R5	2.2k 1%	1	0.24 / 25	Mic supply
C1	10 uF electrolytic	1	3.99 / 10	Mic coupling
ENC	3D-printed PLA+ enclosure, top and bottom	1	printed in house	
HW1	M2x4 socket head screw	22	7.19 / 100	
HW2	M2 flat washer	22	7.99 / 500	
CLIP	Belt clip	1	14.95	Pack size unverified

Source: hardware/BOM.csv. Whether prices stay public, and renumbering refs to match the schematic (R1 to R7, U1 OLED, U2 INA219), are both PENDING decisions.

Pin map

Function	Nano pin	Notes
Microphone	A0	Through TRRS jack and bias network
Battery divider (fallback)	A1	Used only on units without the INA219
I2C SDA / SCL	A4 / A5	OLED (0x3C) and INA219 (0x40), 400 kHz
Buzzer	D3	PWM
STOP / START	D8 / D9	Internal pull-ups, to ground
White LED	D13	PENDING: series resistor

Function	Nano pin	Notes
Free	D2, D4 to D7, D10 to D12, A2, A3, A6, A7	7-segment code removed in v1.7.3

Power path

Battery + goes through the latching switch, then splits: one branch feeds the A1 divider top, the other goes into INA219 Vin+, out of Vin- and into Nano VIN. The Nano's 5 V regulator powers the OLED, INA219 logic and mic bias.

- INA219 Vcc goes to 5 V, never 9 V. Only Vin+ and Vin- see battery voltage.
- No direct wire may remain from the switch to VIN, or the shunt is bypassed and current reads about 0 mA.
- A negative mA reading means Vin+ and Vin- are swapped.
- On USB with the battery off, about 4 V back-feeds onto VIN, so anything under 6.0 V reads as NO BATTERY.

Measured performance

Measure	Value
Battery capacity at 9 V	About 528 mAh (two run-downs: 526 and 529 mAh; label 1300 mAh at 3.7 V)
Current draw	About 33 mA on result pages, 41 mA on BATTERY page, 44 mA in test mode
Runtime	About 16 h per charge (13 h if left on the BATTERY page)
Charge time	About 1 to 2 h over USB-C
Shelf loss	About 5 to 10% per month; nothing drawn with the switch off

Wiring and schematic

The full hole-by-hole wiring lives in WIRING.md and the KiCad schematic in hardware/schematic. The Nano straddles columns C and G in rows 16 to 30 (VIN at G16). The battery divider sits in rows 1 and 2, the mic bias network in rows 12 to 14, and the

OLED header in I5 to I8.

PENDING wiring details: the power switch and INA219 Vin+ board holes; whether A0 taps the mic node (row 12, as built) or the 100k/90k bias node (as drawn in KiCad); whether row 13 F to J goes to GND; where the GND wire from B19 goes; and whether the white LED has a resistor. The manual also lists the battery divider as 10k / 6.8k while WIRING.md and the BOM say 10k / 6.9k (2.2k + 4.7k), so that should be reconciled too.

Firmware

Firmware v1.7.5 is final: one Arduino sketch of about 1,760 lines that uses about 85% of the Nano's flash and leaves about 370 bytes of RAM free at idle. The same build runs on units with or without the INA219, and updating keeps the saved threshold, last string and learned battery capacity.

How the code works

- **Shot detection:** A0 is sampled at about 4 kHz in 5 ms windows of 20 samples. A window whose peak-to-peak value crosses the threshold on a rising edge, at least 100 ms after the last shot, counts as a shot.
- **Timing:** each split is stored as a 16-bit millisecond value (120 bytes for 60 shots), capped at 65,530 ms so it never wraps; a capped gap shows as 65.53+. TOTALS and AVERAGE come from a 32-bit running sum, so strings have no overall length limit.
- **Random delay:** seeded from the moment START is released, so it doesn't repeat between power cycles.
- **Battery gauge:** INA219 current is read every 0.5 s on every screen and integrated into mAh used. The count is saved every 1 mAh and at the end of each string, rotated over 16 EEPROM slots. Capacity starts at 500 mAh and is learned from full discharges, blending 3:1 toward the newest.
- **Fallback:** if no INA219 answers at boot, the timer reads the A1 divider and shows only OK, LOW or CHARGE.
- **Lean by design:** integer-only math, all text in flash via F(), direct EEPROM reads for saved splits, direct INA219 register access (no library), every timer non-blocking, I2C at 400 kHz, serial logging compiled out by default.

Memory map (EEPROM)

Bytes	Contents
0 to 7	Signature, sound threshold, saved shot count
8 to 127	Last string: up to 60 splits (ms)
128 to 135	Gauge signature, learned capacity, gauge flags
136 to 263	16-slot wear-levelled ring of charge-used records

A factory reset clears bytes 0 to 127 only.

Flashing

Install the Arduino IDE and the Adafruit SSD1306 and Adafruit GFX libraries, open `firmware/ShotTimerMk1_v1_7_5/ShotTimerMk1_v1_7_5.ino`, choose Arduino Nano (clones may need ATmega328P Old Bootloader), and upload over a data USB cable with the battery switched off. A chirp from the buzzer on connect is normal.

Version history

Version	What changed
v1.7.5 (final)	Gaps over 65.5 s show as 65.53+ instead of wrapping; CHARGE screen shows the percentage
v1.7.4	Gauge saved every 1 mAh and at string end; 0.5 mAh added back at power-up; MARK FULL? bounce fix; leaving test mode no longer starts a string
v1.7.3	Correct TOTALS/AVERAGE for long strings; corner battery gauge; MARK FULL? confirm; quieter off-page chirps; one-tap exits; NO BATTERY on USB; serial logging off (+157 bytes RAM); 7-segment code removed
v1.7.2	Learns batteries that hold 9 V to the end; LOW at 15%, CHARGE at 5%; warnings on every idle page

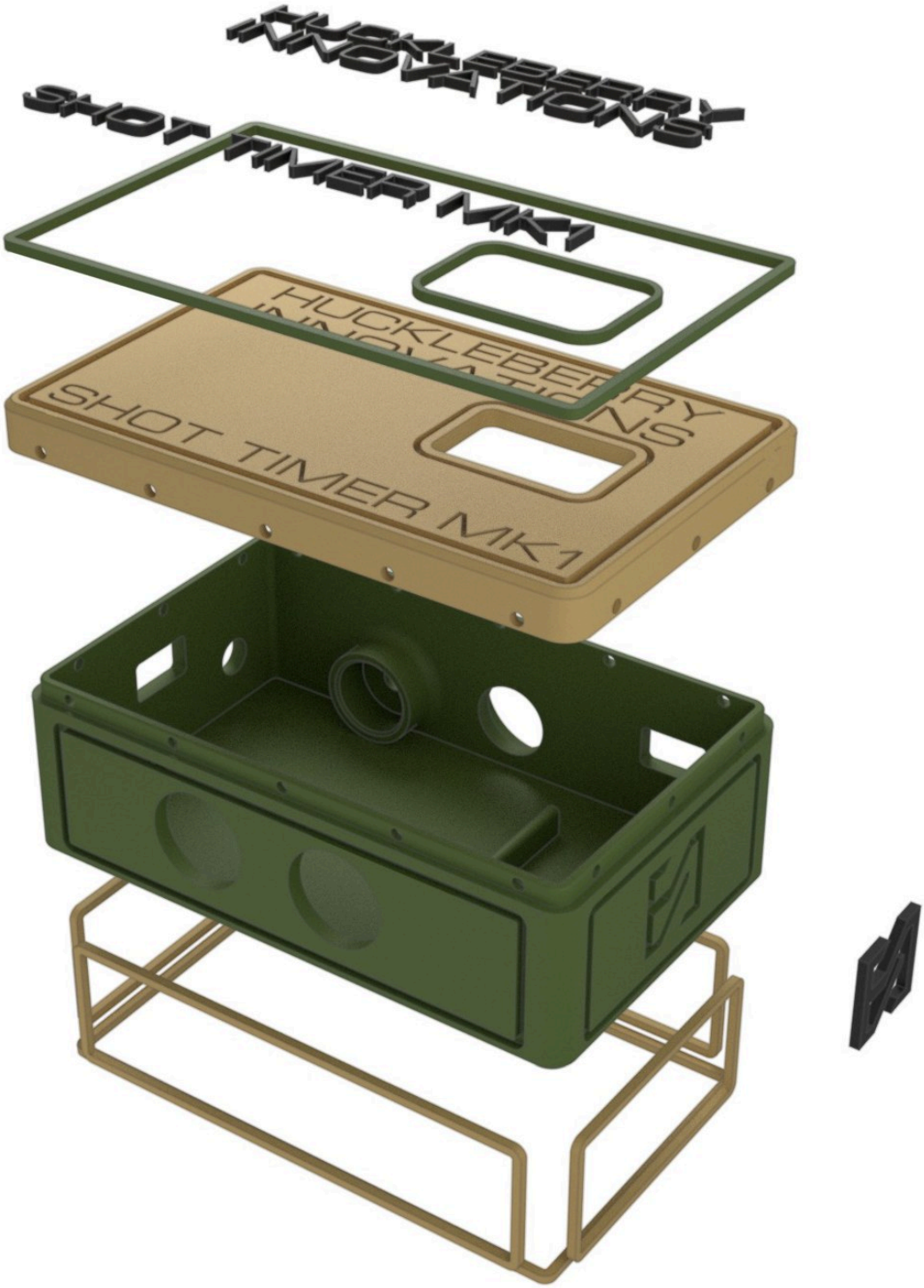
Version	What changed
v1.7.1	Coulomb-counting gauge with percentage, FULL mark and self-calibration
v1.7.0	INA219 battery sensing with live volts and mA; automatic fallback to the A1 divider
v1.6.4	Single LOW beep every 10 s on the BATTERY page (last pre-INA219 version)
v1.6.3	Two-stage warning (LOW 8.5 V, CHARGE 7.9 V)
v1.6.1 to v1.6.2	Live BATTERY refresh; 8-sample battery averaging
v1.6.0	Last string and threshold saved to EEPROM; factory reset; fixed a stack overflow reboot at about 11 shots
v1.5.x	1ST SHOT and AVG SPLIT screens; two-color layout

Source for v1.6.4 and v1.7.0 to v1.7.5 is kept in the source zip; only v1.7.5 is in the repo.

Enclosure

Enclosure v6 is two multi-color PLA+ prints, an olive green bottom shell and a tan lid, measuring 97.0 x 66.0 x 35.0 mm overall with 2.0 mm walls.


HUCKLEBERRY
INNOVATIONS
SHOT TIMER MK1



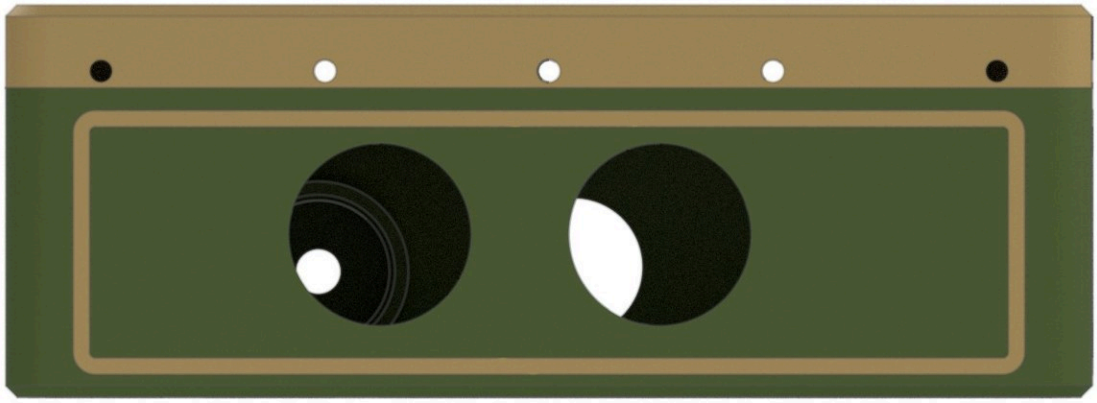
Exploded view: each part prints as one multi-material job; the color bodies are pulled apart to show what each filament prints.

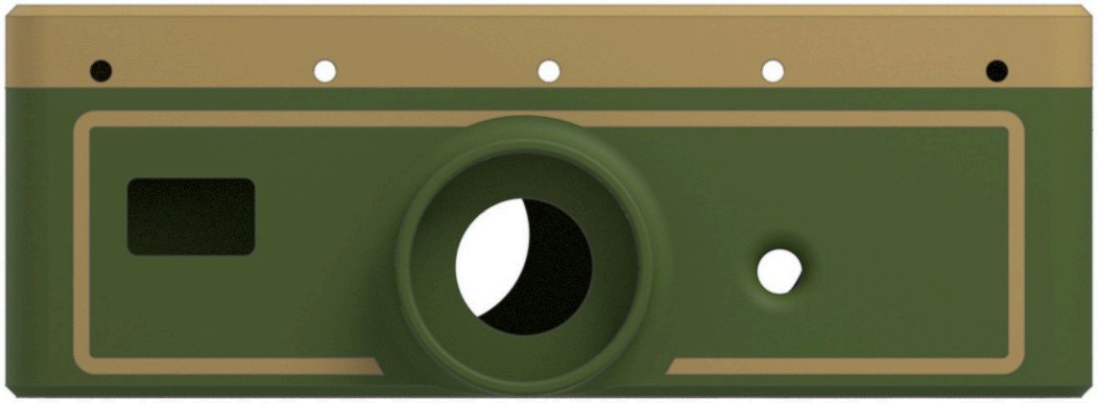
Part	Base color	Inlays	Files
Bottom shell	Olive green	Tan accent lines, black logo	v6 bottom olive green.stl, bottom accents tan.stl, bottom logo black.stl, shot-timer-mk1-bottom.3mf
Lid	Tan	Olive green accent lines, black "Huckleberry Innovations / Shot Timer MK1"	v6 top tan.stl, top accents olive green.stl, top names black.stl, shot-timer-mk1-top.3mf

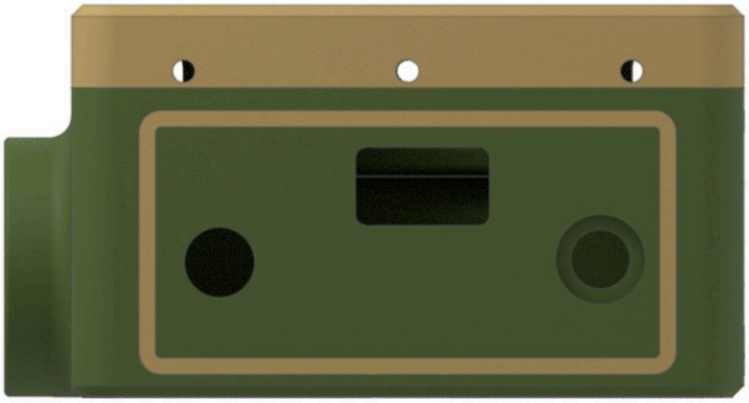
All STL files share one coordinate system, so loading them together in a slicer puts every color body in place. The lid's lip overlaps the shell walls by about 4.5 mm and is held by 16 M2x4 socket-head screws with washers (five along each long side, three at each end).

Openings

Feature	Size (mm)	Location	Purpose
START / STOP holes (2)	16.3 dia	Back wall	16 mm buttons
Power switch hole	12.3 dia	Front wall, inside the raised guard	Latching lit switch
Front wall openings	7.0 x 11.5 slot; 4.5 dia hole	Either side of the switch	PENDING
Port-end openings	7.0 x 12.0 slot; 8.0 and 6.3 dia holes	Right end wall	PENDING (USB-C charging, mic jack, LED?)
Floor holes (2)	6.4 dia	Floor	PENDING (belt clip?)
Screen window	25.0 x 15.2 opening, 30.0 x 20.2 frame	Lid	0.96 in OLED





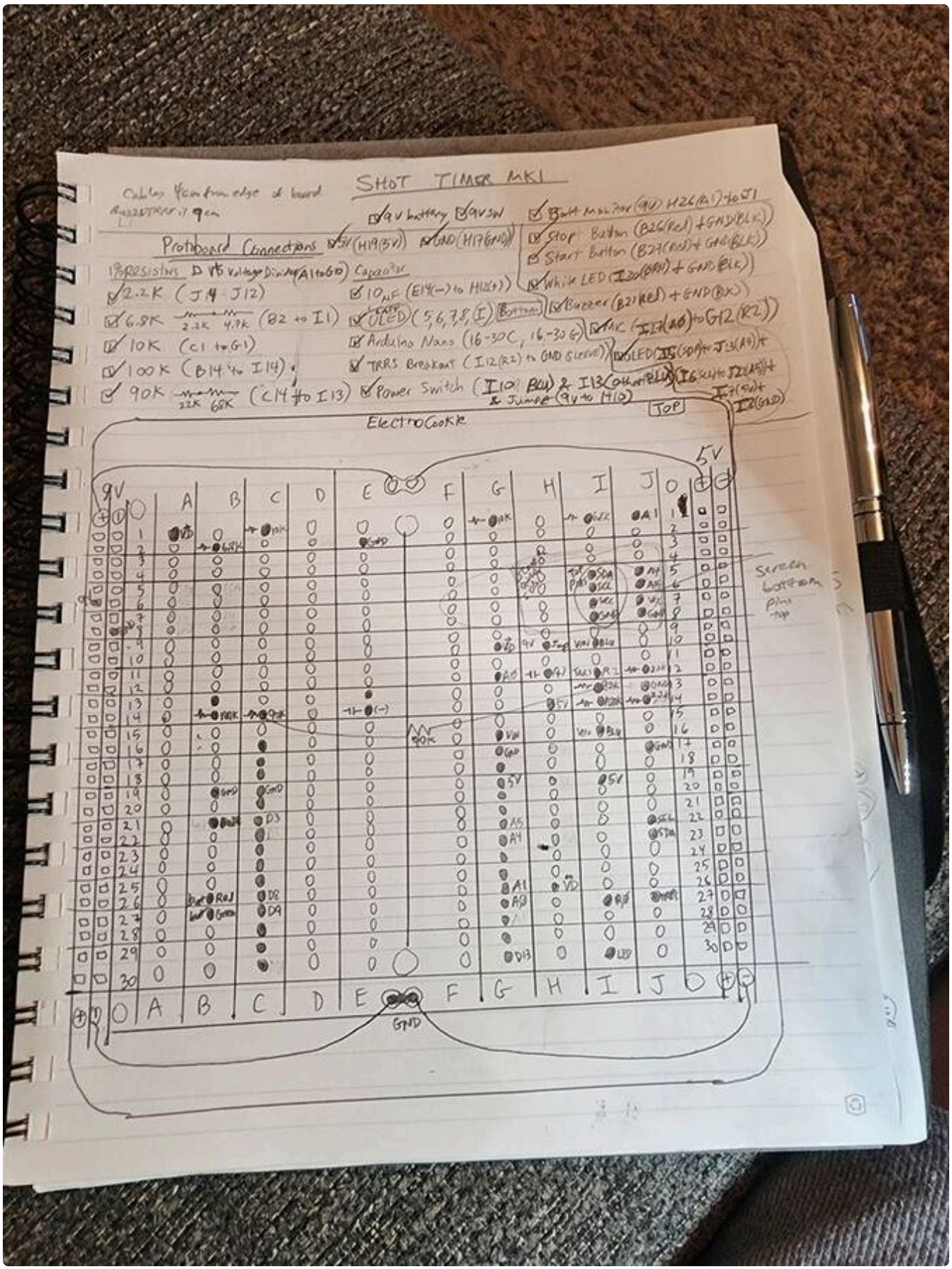


Back, front and right-end views. Whether the switch side is the "front" is itself PENDING.

PENDING: print settings (layer height, infill, supports, orientation, nozzle temperature) are not recorded yet. The BOM lists 22 screws and washers per timer against 16 lid screws in the drawings; the other 6 are presumably for the board, but that is inferred, not confirmed.

Building one

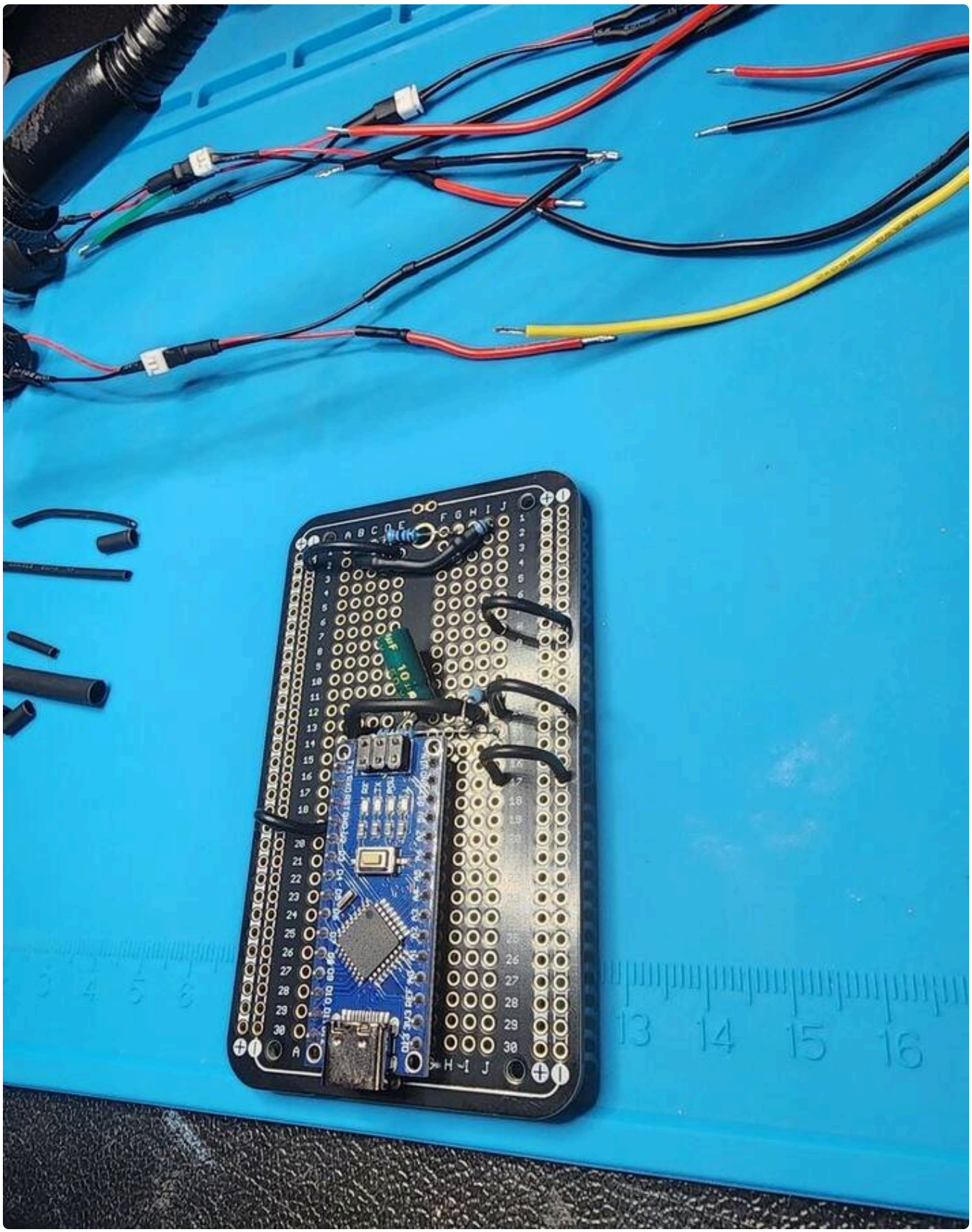
A unit is built in 14 soldering steps, a checkout test and final assembly, all detailed with photos in the Hardware Build Guide. The build photos come from an earlier unit (firmware v1.4.5, before the INA219), so where a photo and the wiring table disagree, the table wins.

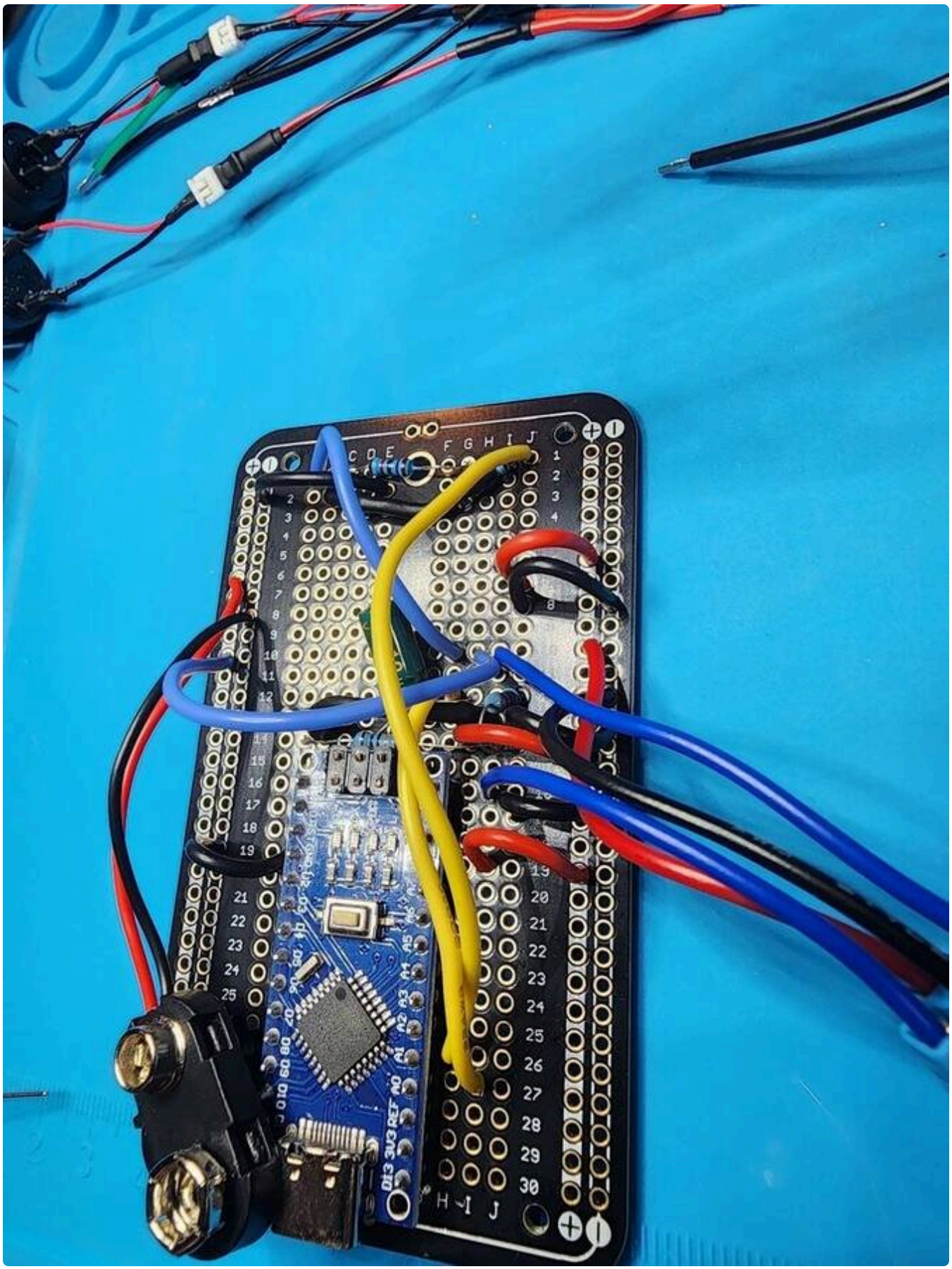

SHOT TIMER MK1
Protboard Connections
ElectroCookie
TOP
GND
5V
9V

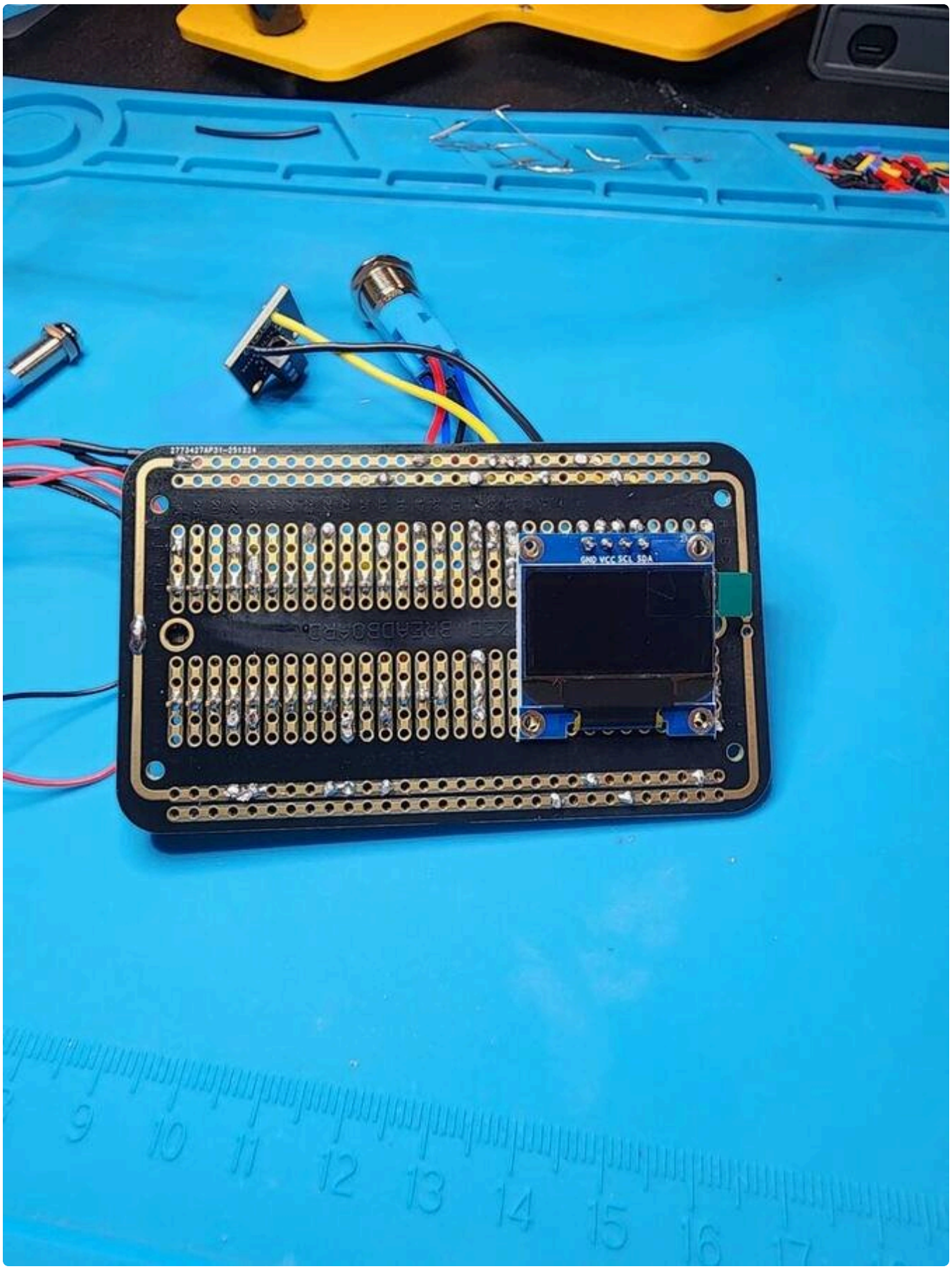
The original hand-drawn notebook layout that WIRING.md was transcribed from.

Build order

1. Make the interconnect cabling: lead pairs for every panel part, 2-pin connectors, heat-shrink. (Lead lengths of 4 cm, 9 cm for the buzzer: PENDING.)
2. Install the Nano, resistors, capacitor, 5 V and ground links.
3. Add the analog signal jumpers (A1 sense, A0 mic).
4. Install the 9 V battery clip.
5. Install the power switch and its LED (exact holes PENDING).
6. Install the INA219, insulated with Kapton or heat-shrink (no photo yet: PENDING).
7. Run the OLED wires.
8. Install the TRRS breakout.
9. Wire the white LED.
10. Wire the buzzer.
11. Wire the START button.
12. Wire the STOP button.
13. Checkpoint: inspect joints, check for shorts with a multimeter.
14. Solder the OLED last, on the underside of the board.






GND VCC SCL SDA

Steps 2, 7 and 14: Nano and passives, panel wiring, OLED fitted.

Checkout test

- [ ] With the switch off, no short between 5 V and GND or 9 V switched and GND
- [ ] Flash v1.7.5 over USB with the battery off
- [ ] Start screen shows Shot Timer Mk1, v1.7.5 and the battery gauge
- [ ] Test mode (hold START 2 s) shows BATT:INA219 and RAM about 370; PEAK jumps on a clap
- [ ] BATTERY page on battery shows about 9 V and a positive current of about 41 mA
- [ ] Ten claps register as 10 shots at threshold 150
- [ ] Charge, then mark FULL on the BATTERY page
- [ ] Factory reset before shipping

Final assembly and packaging

Mount the buttons, switch, LED and jack in the shell and plug their leads into the board. Fix the insulated INA219 with foam tape, line the OLED up behind the screen window, and route the I2C wires along the walls with slack, because a pinched SDA wire leaves the screen dark. Fit the battery, close the lid with the M2 screws and washers, and add the belt clip.


9V



Image — Packed in the Huckleberry Innovations box

Final assembly, the finished unit powered on, and the retail box.

Documents and where they live

The finished documents live in the GitHub repo; the sources that build them live in shot-timer-mk1-source.zip in the project files. Edit the source and rebuild rather than editing a PDF, and keep the manual, quick-start card, build guide, BOM, WIRING.md, schematic and enclosure drawings consistent with each other.

Deliverable	In the repo	Built from (in the zip)	Rebuild
User manual v1.7.5	docs/Shot_Timer_Mk1_Manual_v1.7.5.pdf	manual/manual.html, scr175b/ screen images	render_pdf.py; screens.py regenerates the OLED images
Quick-start fold card	docs/Shot_Timer_MK1_Quick_Start_Fold_v1.7.5.pdf	quickstart-card/fold.html	render_pdf.py
Hardware build guide	docs/Shot_Timer_Mk1_Hardware_Build_Guide.pdf	hardware-guide/guide.py, board.py, photos/	board.py, guide.py, render_pdf.py (parts table read from BOM.csv)
Schematic	hardware/schematic/ (KiCad + PDF)	kicad/gen.py	gen.py, or edit in KiCad 7+
Enclosure drawings	hardware/enclosure/…Drawings.pdf	enclosure/doc.py, render.py (Blender), renders_jpg/	doc.py, pdf.py
Enclosure print files	hardware/enclosure/ (6 STL, 2 3MF)	not in the zip	from CAD
Parts list	hardware/BOM.csv	github-repo/ snapshot	edit directly
Wiring	hardware/WIRING.md	github-repo/ snapshot	edit directly
Firmware	firmware/ShotTimerMk1_v1_7_5/	firmware/ (plus v1.6.4, v1.7.0 to v1.7.4)	Arduino IDE

Tools for rebuilding: Python 3 with Playwright and Pillow, Blender 4.x (only to re-render the enclosure), KiCad 7 or newer, and the Inter and Lora fonts. The repo has no automated build check.

The repo's two commits are "Shot Timer MK1 v1.7.5: firmware, manuals, licenses" (2026-10-18) and "Add hardware documentation, KiCad schematic and enclosure files" (2026-10-19). The hardware documents are a first release and carry DRAFT notes and question marks where details are unconfirmed.

History and decisions

The timer went from a voltage-divider battery check to a learned coulomb-counting gauge in October 2026, and was published as open source on 2026-10-18 and 2026-10-19. Exact dates for versions before v1.7.0 aren't recorded in the sources.

Date	Event
2026-10-19	Hardware docs, KiCad schematic and enclosure files pushed to GitHub; hardware feedback list opened (16 items)
2026-10-18	Firmware v1.7.5, manuals and licenses pushed to GitHub; manual v1.7.5 dated
2026-10-18	Second full battery run-down on v1.7.4: 529 mAh, learned capacity blended to 528 mAh
2026-10-16	INA219 added high-side in the battery line (firmware v1.7.0 onward uses it)
Oct 2026	Firmware v1.7.0 to v1.7.5 developed (gauge, warnings, long-gap fix)
Earlier	v1.5.x to v1.6.4: two-color screens, EEPROM save, factory reset, divider-based LOW/CHARGE warnings
Earlier	Build photographed on firmware v1.4.5, before the INA219

Decisions already made

Decision	Why
Rechargeable 9 V USB-C lithium battery only	It holds about 9.1 V until empty, so alkaline packs would trip false LOW/CHARGE warnings
Gauge by coulomb counting, not voltage	Voltage gives no warning on a lithium pack; the count does
INA219 driven by direct register access	Saves flash and RAM on the Nano

Decision	Why
Same firmware with or without INA219	Early divider-only units keep working
Keep the power-up buzzer chirp	Handy cue that USB is live; harmless
Open source under GPL-3.0+, CERN-OHL-S-2.0 and CC BY-SA 4.0	Others may build, modify and sell units if they share changes
Brand names excluded from the licenses	Derivatives must use their own name

Open items

Nothing on this list blocks using the timer; these are 18 details to confirm so the hardware documents can drop their DRAFT marks. Items 1 to 16 come from the hardware feedback sheet and are all still open as of 2026-10-19; A and B are two inconsistencies found while assembling this book. The five High items change the wiring diagram and schematic, so they are the most useful to answer first, mostly by opening a built timer.

#	Question	Priority	Documents affected	Status
1	Power switch wiring: which board holes the switch input, output and LED wires use, and where INA219 Vin+ connects	High	Build guide, WIRING.md, board diagram, schematic	Open
2	Mic circuit: does A0 tap the mic node (row 12, as built) or the 100k/90k bias node (row 14, as drawn in KiCad)?	High	Build guide, WIRING.md, board diagram, schematic, manual	Open
3	Row 13 F to J: connected to GND? (bottom end of the 90k resistor)	High	WIRING.md, board diagram	Open
4	Row 19: where does the GND wire from B19 go?	High	WIRING.md, board diagram	Open
5	White LED: is there a resistor anywhere? (manual says "through 1k", board shows none)	High	All hardware docs, manual	Open

#	Question	Priority	Documents affected	Status
6	Wire lengths: are 4 cm from the board edge and 9 cm for the buzzer still right?	Normal	Build guide	Open
7	INA219 photo installed (next build is fine)	Normal	Build guide, step 6	Open
8	Opening functions: switch-side 7 x 11.5 slot and 4.5 hole; port-end 7 x 12 slot, 8.0 and 6.3 holes (USB-C charging, mic jack, LED?)	Normal	Enclosure drawings	Open
9	Floor holes (6.4 dia, x2): belt clip mounting?	Normal	Enclosure drawings	Open
10	Is the switch side the "front"?	Normal	Enclosure drawings	Open
11	Print settings: layer height, infill, supports, orientation, nozzle temp	Normal	Enclosure drawings	Open
12	Battery: is the $25 for one battery or a pack?	Normal	BOM	Open
13	Prices: keep them public in BOM.csv or remove the column?	Normal	BOM	Open
14	Copyright name: "Josue F. Kure / Huckleberry Innovations LLC" or the LLC only?	Normal	All documents, firmware header	Open
15	Renumber BOM, WIRING.md and build guide refs to match the schematic (R1 to R7, U1 OLED, U2 INA219)?	Normal	BOM, WIRING.md, build guide	Open
16	Run the KiCad 9 Electrical Rules Checker and note any errors	Normal	Schematic	Open

#	Question	Priority	Documents affected	Status
A	Battery divider bottom: the manual says 6.8k, WIRING.md and the BOM say 6.9k (2.2k + 4.7k). Which is right?	Normal	Manual, WIRING.md, BOM	Open
B	Screws: the BOM lists 22 M2 screws and washers, the drawings show 16 for the lid. What are the other 6 for?	Normal	BOM, enclosure drawings	Open

Answers go to the Apply hardware doc feedback thread, which updates and rebuilds the documents. Item 16 needs KiCad 9 on your own computer.